DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are "not alive" (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed `.qmd` file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```r
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

# Load the MA_schools dataset
data(MA_schools)
```

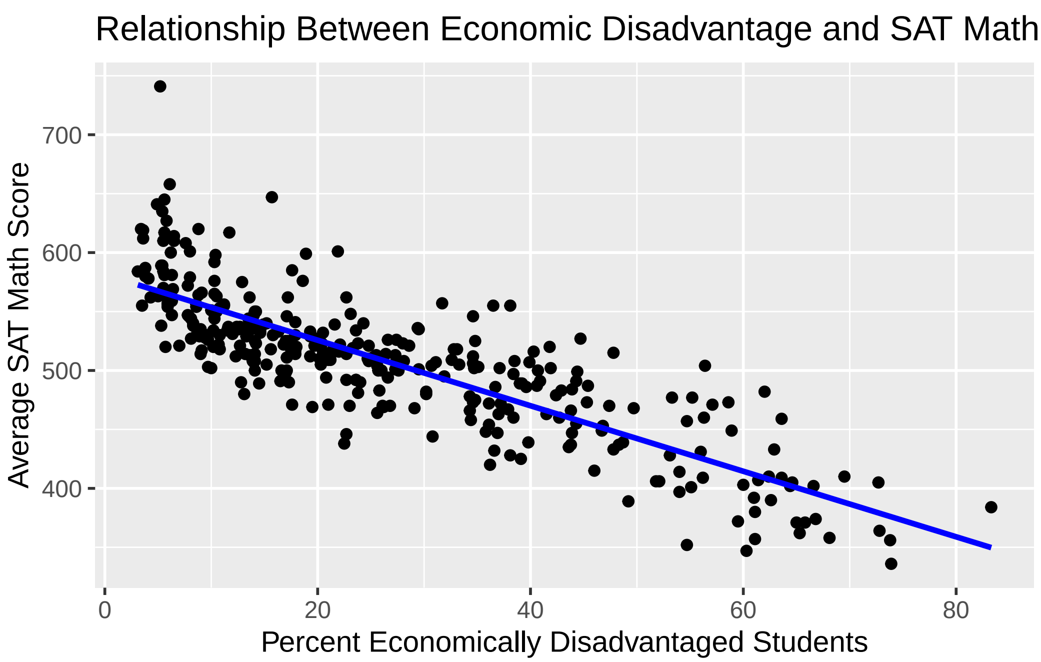
Data Exploration (3 points)

1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

```
# YOUR CODE HERE
ggplot(data = MA_schools, aes(x = perc_disadvan, y = average_sat_math)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, color = "blue") +
  labs(
    title = "Relationship Between Economic Disadvantage and SAT Math Scores",
    x = "Percent Economically Disadvantaged Students",
    y = "Average SAT Math Score"
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Answer: scatterplot shows a negative relationship between 'perc_disadvan'and 'average_sat_math'. As Percent Economically Disadvantaged Students increase, the Average SAT Math Score will decrease.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
# YOUR CODE HERE
model1 <- lm(average_sat_math ~ perc_disadvan, data = MA_schools)
summary(model1)
```

```
Call:
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)

Residuals:
   Min     1Q Median     3Q    Max 
-80.74 -21.26  -4.12  18.54 174.17 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)   581.2811     3.2668   177.9   <2e-16 ***
perc_disadvan  -2.7798     0.1011   -27.5   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom
Multiple R-squared:  0.6962,	Adjusted R-squared:  0.6953 
F-statistic: 756.2 on 1 and 330 DF,  p-value: < 2.2e-16
```

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
# YOUR CODE HERE
sat_effects <- coef(model1)
sat_effects
```

```
  (Intercept) perc_disadvan 
   581.281054     -2.779787 
```

Intercept: 581.281054.

Slope: -2.779787.

3 (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: For every 1 point increase in 'proportion of economically disadvantaged students', 'average SAT math score' decreases by an average of 2.78 points.

Making Predictions (8 points)

1 (4 points)

Use `model1` to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
# YOUR CODE HERE
new_data <- data.frame(perc_disadvan = 50)
predicted_score <- predict(model1, newdata = new_data)
predicted_score
```

```
       1
442.2917
```

Predicted average SAT math score: 442.2917.

2 (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
# YOUR CODE HERE
new_data <- data.frame(perc_disadvan = 50)
predict(model1, newdata = new_data, interval = "prediction", level = 0.95)
```

```
       fit      lwr      upr
1 442.2917 376.0542 508.5292
```

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```r
# YOUR CODE HERE
model2 <- lm(average_sat_math ~ perc_disadvan + size, data = MA_schools)
summary(model2)
```

```

Call:
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)

Residuals:
    Min      1Q  Median      3Q     Max 
-81.342 -20.802  -3.872  18.211 167.251 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)   588.1898     7.6067  77.325   <2e-16 ***
perc_disadvan  -2.7772     0.1063 -26.120   <2e-16 ***
sizemedium    -11.9134     7.5346  -1.581    0.115    
sizelarge      -6.3615     6.9231  -0.919    0.359    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.5 on 328 degrees of freedom
Multiple R-squared:  0.6988,	Adjusted R-squared:  0.696 
F-statistic: 253.6 on 3 and 328 DF,  p-value: < 2.2e-16
```

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```r
# YOUR CODE HERE
summary(model1)$r.squared
```

```
[1] 0.6961948
```

```r
summary(model2)$r.squared
```

```
[1] 0.6987587
```

```r
c(Model1_R2 = summary(model1)$r.squared,
  Model2_R2 = summary(model2)$r.squared)
```

```
Model1_R2 Model2_R2
0.6961948 0.6987587
```

Answer: Model 2 explains slightly more variance [$R^2 = 0.6988$(model2)$>$ 0.6962(model1)], but the improvement is minimal.

3 (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```r
anova(model1, model2)
```

```
Analysis of Variance Table

Model 1: average_sat_math ~ perc_disadvan
Model 2: average_sat_math ~ perc_disadvan + size
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    330 371191
2    328 368058  2    3132.6 1.3958 0.2491
```

Conclusion: p-value = 0.2491 (from the ANOVA test) $>$ 0.05 means: Fail to reject H . adding 'size' does not significantly improve model fit.